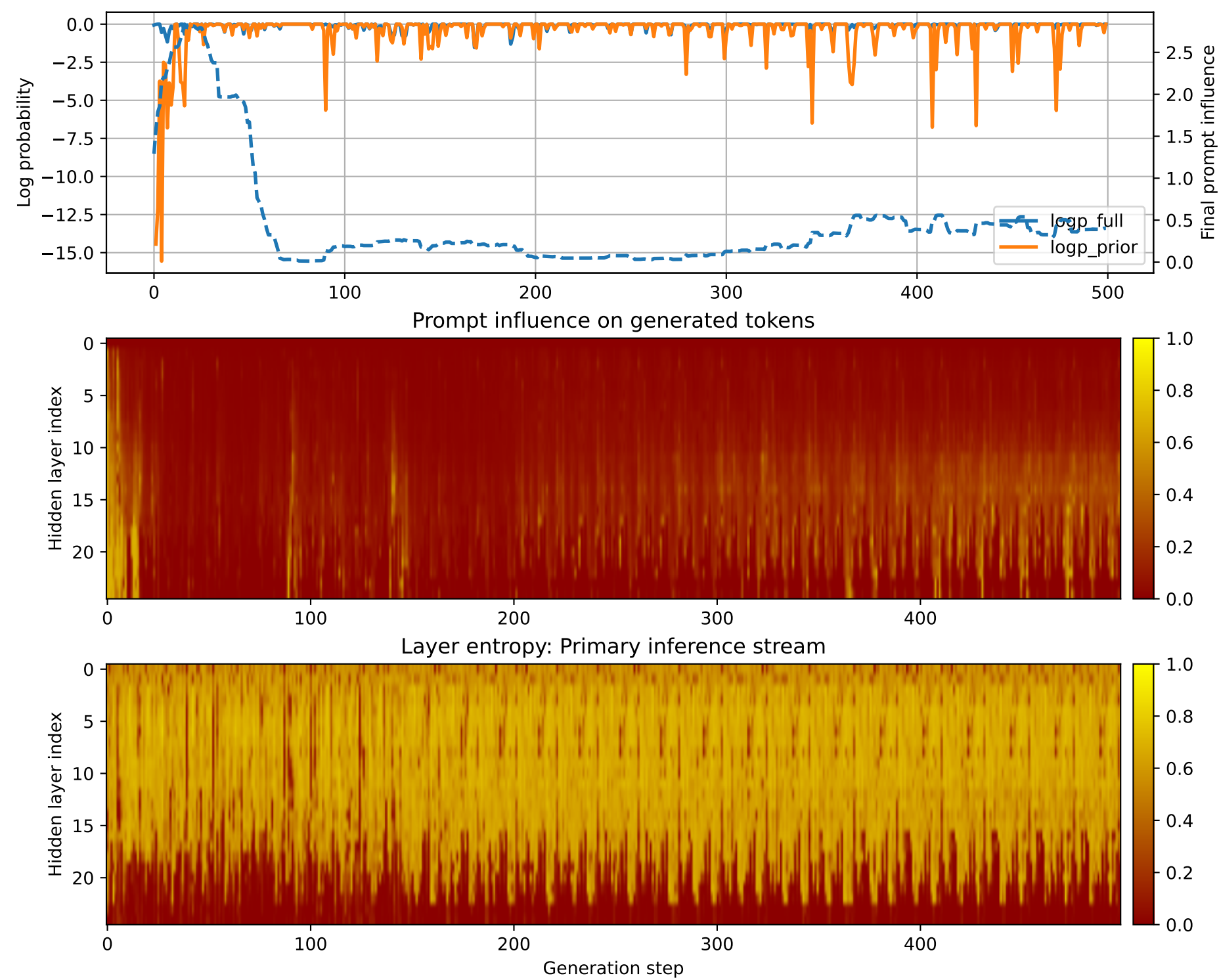


Prompt 1



PROMPT:
Task ID: PYC038 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC038 Your code must contain this required symbol:
def lcm Implement this task: define
function lcm(a,b) using gcd.. Start your answer with the
check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: lcm # Define function lcm(a, b) using gcd() def
lcm(a, b): """ Calculate the least
common multiple (LCM) of two integers a and b using the greatest
common divisor (GCD) of a and b. Args:
a (int): The first integer. b (int): The second integer.
Returns: int: The LCM of a and
b. """ # Calculate the GCD of a and b gcd = gcd(a, b)
Calculate the LCM by dividing the GCD
by the larger number lcm = gcd // max(a, b) return lcm
Example usage: print(lcm(12, 18)) #
Output: 36 print(lcm(18, 24)) # Output: 72 print(lcm(24, 36)) #
Output: 72 print(lcm(36, 48)) # Output: 144
print(lcm(48, 60)) # Output: 288 print(lcm(60, 72)) # Output:
432 print(lcm(72, 96)) # Output: 144
print(lcm(96, 120)) # Output: 288 print(lcm(120, 144)) #
Output: 432 print(lcm(144, 168)) # Output: 72
print(lcm(168, 252)) # Output: 144 print(lcm(252, 300)) #
Output: 72 print(lcm(300, 360)) # Output: 144
print(lcm(360, 480)) # Output: 72 print(lcm(480, 600)) #
Output: 144 print(lcm(600, 720)) # Output: 144
print(lcm(720, 960)) # Output: 144 print(l